LifeTag: Smart Emergency Health ID System using IoT

An Engineering Project in Community Service

Phase-I Report

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Bonafide Certificate

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This project report (Phase I) is submitted for the Project Viva-Voce examination held on ……..

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We, hereby declare that this report entitled **"LifeTag: Smart Emergency Health ID System using IoT"** represents our original work carried out for the EPICS project as students of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

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ABSTRACT

The LifeTag system proposes a Smart Emergency Health ID solution that enables quick access to essential patient medical information during emergencies. Each individual is assigned a unique QR-based Health ID containing critical medical data such as blood group, allergies, and past illnesses. The system integrates IoT sensors to monitor vital parameters such as heart rate and body temperature and store the data in a cloud database for real-time access.

Phase-I focuses on system design, literature review, architecture planning, and module identification. Approximately 60% of the project work has been completed. The remaining work involves implementation, integration, and testing.

1. INTRODUCTION

Healthcare emergencies often require immediate access to patient medical history. In many situations, patients are unable to communicate their medical details due to unconsciousness, trauma, panic, or stress. This delay in obtaining medical history can significantly affect treatment decisions, increase medical risks, and reduce the efficiency of emergency healthcare services.

The **LifeTag: Smart Emergency Health ID System using IoT** is designed to address this problem by combining QR-based patient identification, IoT health monitoring, and cloud-based medical record storage into a single integrated healthcare solution. The system focuses on providing quick, reliable, and secure access to essential medical information during emergency situations.

The LifeTag system provides the following key features:

• instant access to critical medical data
• real-time monitoring of vital signs
• offline emergency information availability
• patient-controlled data access

In the proposed system, every individual is assigned a **unique QR-based Health ID** that stores important medical information such as blood group, allergies, existing medical conditions, and emergency contacts. During emergencies, healthcare professionals can scan the QR code to retrieve essential medical details immediately, allowing treatment to begin without delay.

The system also integrates **IoT sensors** that continuously monitor vital parameters such as heart rate and body temperature. These sensors transmit data to a **cloud database**, where medical records can be stored and accessed in real time. This ensures that patient health information remains updated and available when required. Another important

feature of the system is **offline emergency access**, which allows basic patient information to be retrieved even when internet connectivity is unavailable. This makes the system particularly useful in **rural healthcare environments and emergency scenarios** where network access may be limited.

By integrating identification, monitoring, and data storage technologies into one platform, the LifeTag system aims to improve emergency healthcare response time, support doctors in decision-making, and enhance patient safety.

1.1 Motivation

Many hospitals maintain digital medical records, but these systems are often not interconnected or easily accessible during emergencies. When patients are taken to a hospital where their records are not available, doctors may need time to collect important medical information before beginning treatment. This delay can affect emergency response and patient safety. Rural healthcare environments face additional challenges due to limited infrastructure and unreliable internet connectivity.

Situations like the **COVID-19 pandemic** highlighted the importance of quickly accessible medical data. During COVID-19, many patients had to rush to unfamiliar hospitals without access to their family doctor or known healthcare personnel. Hospitals often required time to collect documentation and verify patient medical history before beginning treatment.

If even a **basic summary of critical medical information** — such as blood group, allergies, existing medical conditions, or emergency contacts — is available immediately, treatment can begin faster while detailed verification can be completed later. Quick access to this information can significantly improve emergency healthcare response.

Another motivating factor is the growing use of **IoT-based health monitoring devices**, which allow continuous tracking of vital signs. However, these systems are usually not connected with emergency identification or hospital record systems. Integrating identification, monitoring, and medical record storage can make healthcare systems more efficient and reliable.

These challenges motivated the development of a system that:

• provides instant emergency medical access
• integrates IoT monitoring with identification
• supports offline emergency access
• enables patient-controlled data sharing
• works in rural healthcare environments
• remains low-cost and scalable

The **LifeTag system** is designed to address these needs by combining QR-based identification, IoT health monitoring, and cloud-based medical record storage into one unified healthcare solution.

1.2 Objective

The primary objective of the **LifeTag: Smart Emergency Health ID System using IoT** project is to design and develop a smart healthcare identification and monitoring system that enables quick access to essential patient medical information during emergency situations. The system aims to reduce the delay in treatment caused by the unavailability of medical history and improve decision-making efficiency for healthcare professionals.

One of the main objectives of this project is to design a **QR-based Health ID system** that uniquely identifies each patient and stores critical medical information such as blood group, allergies, chronic diseases, and emergency contact details. The QR-based approach ensures fast identification and minimal infrastructure requirements.

Another important objective is to **implement IoT-based vital monitoring** using sensors capable of measuring physiological parameters such as heart rate and body temperature. These sensors continuously monitor patient health data and send it to the system for storage and analysis.

The project also aims to **develop a cloud-based medical data storage system** that enables secure storage and retrieval of patient information. Cloud storage ensures accessibility, scalability, and reliability of healthcare data.

A key objective is to **enable offline emergency access**, allowing doctors to access essential patient information even in environments where internet connectivity is unavailable. This feature is particularly useful in rural healthcare settings.

The system is also designed to **ensure patient-controlled data access**, allowing individuals to control which medical information is shared with healthcare providers while still enabling emergency access to critical data.

Finally, the project aims to **develop a working prototype** that demonstrates the integration of QR identification, IoT monitoring, cloud storage, and emergency access into a unified healthcare solution.

2. EXISTING WORK / LITERATURE REVIEW

Healthcare technology has evolved significantly with the introduction of Electronic Health Records (EHR), Internet of Things (IoT) devices, cloud computing, and digital identification systems. These technologies have improved medical data storage,

monitoring, and accessibility, enabling healthcare providers to manage patient information more efficiently. However, integration between these technologies in emergency healthcare systems remains limited.

Electronic Health Record (EHR) systems allow hospitals to store patient medical data digitally, reducing reliance on paper documentation and improving record management. EHR systems help healthcare professionals maintain patient history, prescriptions, diagnostic reports, and treatment records in an organized manner. Despite these advantages, most EHR systems are **hospital-specific and centralized**, meaning patient data stored in one healthcare institution is not always accessible in another.

In emergency situations, this lack of interoperability becomes a major challenge. When patients are admitted to unfamiliar hospitals, doctors may not have access to their previous medical records. As a result, healthcare providers must rely on patient communication, family members, or repeated diagnostic procedures, which can delay treatment and increase healthcare costs. This highlights the need for a system that allows **quick and universal access to essential medical information** during emergencies.

IoT-based healthcare monitoring systems have gained attention due to their ability to continuously track patient vital signs using wearable devices and sensors. These systems enable real-time monitoring of physiological parameters such as heart rate, body temperature, blood pressure, and oxygen levels. Continuous monitoring helps detect health abnormalities early and improves patient care in both hospital and home environments.

Despite the benefits of IoT healthcare monitoring, most existing systems operate as **independent monitoring solutions**. They focus primarily on data collection and visualization but do not integrate with emergency patient identification systems or medical record databases. This limits their usefulness in emergency scenarios where both identification and monitoring information are required simultaneously.

Cloud computing has also become an important technology in healthcare data management. Cloud-based healthcare systems allow remote storage and retrieval of patient records, making data accessible from multiple locations. Cloud platforms provide scalability, reliability, and centralized management of healthcare information. Hospitals and healthcare providers can use cloud infrastructure to store large volumes of patient data securely.

However, cloud-based healthcare systems depend heavily on **stable internet connectivity**, which may not always be available in emergency situations or rural healthcare environments. In such conditions, accessing medical records stored only in the cloud becomes difficult. Therefore, systems that combine cloud storage with **offline emergency access capabilities** can provide more reliable solutions.

Another important technology used in identification systems is the **Quick Response (QR) code**. QR codes are widely used in authentication, ticketing, inventory

management, and digital payments because they are easy to generate, store information efficiently, and can be scanned quickly using mobile devices. QR-based identification systems are low-cost and require minimal infrastructure.

Although QR codes are commonly used for identification in many industries, their application in emergency healthcare systems remains limited. Integrating QR-based identification with medical records and monitoring systems can significantly improve the speed and efficiency of emergency healthcare services.

Research studies in smart healthcare systems have explored technologies such as wireless sensor networks, wearable medical devices, cloud-based health monitoring platforms, and digital patient identification methods. These studies demonstrate the potential of combining computing technologies with healthcare services to improve patient care and monitoring.

However, most existing research focuses on **individual components rather than integrated systems**. For example, some studies focus only on IoT-based patient monitoring, while others focus on cloud-based electronic health record systems or digital identification technologies. Very few solutions combine identification, monitoring, storage, and emergency access into a single healthcare platform.

The lack of integration between these technologies creates challenges in emergency healthcare scenarios where **fast identification, immediate access to medical history, and real-time monitoring** are all required simultaneously. This gap between existing healthcare technologies and emergency healthcare needs motivates the development of integrated healthcare solutions.

2.1 Research Gap and Proposed Contribution

Based on the review of existing healthcare technologies, it is clear that while EHR systems, IoT monitoring, cloud storage, and QR-based identification are individually well developed, their **integration into a unified emergency healthcare system is still limited**.

The **LifeTag: Smart Emergency Health ID System using IoT** addresses this research gap by combining multiple healthcare technologies into a single platform. The proposed system integrates:

• QR-based emergency patient identification
• IoT real-time health monitoring
• cloud-based medical record storage
• offline emergency access capability
• patient-controlled data sharing

This integrated approach allows healthcare professionals to quickly access essential patient information, monitor vital signs, and retrieve medical records during emergency situations.

Unlike traditional EHR systems that are limited to individual hospitals, the LifeTag system focuses on **portable patient identification and emergency accessibility**. Unlike standalone IoT monitoring systems, the proposed system connects monitoring data with patient identity and medical history. Additionally, offline emergency access ensures system usability in low-connectivity environments.

By combining identification, monitoring, and medical data storage into one platform, the LifeTag system provides a **practical and scalable solution for emergency healthcare scenarios**, particularly in rural healthcare environments where infrastructure limitations exist.

This integration of multiple healthcare technologies represents the key contribution of the proposed system and distinguishes it from existing healthcare monitoring or record-management solutions.

3. TOPIC OF THE WORK

The LifeTag system is designed as an integrated emergency healthcare identification and monitoring platform that combines QR-based identification, IoT health monitoring, and cloud-based medical record storage. The system is modular in nature, allowing each component to operate independently while contributing to the overall functionality of the system.

The main goal of the system is to provide **quick access to essential patient medical information during emergency situations** while supporting continuous health monitoring through IoT devices.

3.1 System Design / Architecture

The LifeTag system architecture consists of five major modules:

- Patient Registration Module
- QR Health ID Module
- IoT Monitoring Module
- Cloud Database Module
- Emergency Access Module

These modules interact through a centralized cloud platform.

System architecture ensures:

- real-time data flow
- secure medical record storage
- emergency accessibility
- modular scalability

Fig. 1 System Architecture Diagram

3.2 Module Description

Patient Registration Module

This module collects patient information such as name, age, blood group, allergies, medical history, and emergency contacts. The collected information is stored in the database and linked to a unique Health ID.

QR Health ID Module

The QR module generates a unique QR code for each patient. The QR code acts as a digital identity that allows quick retrieval of medical information during emergencies.

IoT Monitoring Module

This module includes sensors that monitor patient vital signs such as heart rate and body temperature. Sensor data is transmitted to the microcontroller and uploaded to the cloud database.

Cloud Database Module

The cloud database stores patient medical records and IoT sensor data. It ensures secure storage and remote accessibility of medical information.

Emergency Access Module

This module allows doctors to scan the QR code and access essential patient medical information instantly. Offline emergency data can also be displayed when connectivity is unavailable.

3.3 Working Principle

The system operates in a sequence of steps:

1. Patient registers in system
2. Health ID generated
3. QR code created
4. Medical data stored in database
5. IoT sensors monitor vitals
6. Sensor data uploaded to cloud
7. QR scanned during emergency
8. Emergency data displayed
9. Doctor accesses full record

Fig. 2 System Workflow Diagram

3.4 Data Flow

The data flow in the LifeTag system follows this path:

IoT Sensors → Microcontroller → Cloud Database → QR Health ID → Hospital Scanner → Doctor Interface

The architecture ensures that patient data remains synchronized between monitoring devices and the database.

3.5 Implementation Plan

Phase-II implementation will include:

- sensor interfacing
- microcontroller programming
- QR generation software
- cloud database integration
- testing and validation

Hardware components may include:

- heart rate sensor
- temperature sensor
- microcontroller (Arduino/ESP32)
- QR scanner or mobile application

Software components include:

- database system
- QR generator
- cloud platform
- monitoring dashboard

3.6 Results and Discussion (Phase-I)

Phase-I focused on:

- problem identification
- literature review
- system architecture design
- module planning
- documentation

Project completion status: **~60%**

Table 1 Phase-I Completion Status

Module	Status
Literature Review	Completed
Architecture Design	Completed

Module	Status
Database Design	Completed
QR Module	Completed
IoT Planning	Completed
Documentation	Completed

The Phase-I results confirm the feasibility of integrating QR identification, IoT monitoring, and cloud storage into one healthcare system.

3.7 Proposed Work for Next Phase

Remaining work includes:

- hardware setup
- sensor integration
- database connectivity
- QR scanning implementation
- system testing
- prototype demonstration

The next phase will focus on developing a working prototype of the LifeTag system.

4. Individual Contribution

Digvijay Singh — IoT Sensor Integration

Responsible for sensor selection, IoT communication planning, and hardware integration design. Studied microcontroller-sensor interfacing and contributed to architecture planning for real-time monitoring.

Aryan Walia — Cloud Database Development

Responsible for database schema design, cloud storage planning, and connectivity architecture between system modules.

Lav Singh — QR Health ID Implementation

Responsible for QR code generation module design and emergency identification workflow.

Vidisha Adhiya — Project Ideation, Research, Documentation

Proposed the LifeTag system concept, defined project objectives, conducted literature review, and prepared project documentation.

Riya Garg — Assistance, Testing, Reference Collection

Supported development planning, testing preparation, and research reference collection.

5. Conclusion

The LifeTag system demonstrates how multiple modern healthcare technologies can be integrated into a single emergency-support platform. The project addresses a critical limitation in existing healthcare infrastructure — the lack of immediate access to patient medical information during emergency situations. By combining QR-based identification, IoT health monitoring, and cloud-based medical record storage, the proposed system improves both accessibility and reliability of patient data.

Electronic Health Record systems have already improved hospital data management, but interoperability and emergency accessibility remain major challenges. Similarly, IoT healthcare monitoring systems provide continuous tracking of vital signs, yet they often operate independently without integration into identification or emergency response mechanisms. Cloud storage enables remote access to medical data but depends heavily on network availability. The LifeTag system attempts to bridge these gaps by integrating these technologies into a unified architecture.

The QR Health ID component enables fast identification of patients during emergencies with minimal infrastructure requirements. The IoT monitoring module supports real-time tracking of vital parameters such as heart rate and temperature. The cloud database

ensures secure storage and remote availability of medical records, while the offline emergency access feature improves system reliability in low-connectivity environments.

Phase-I of the project successfully completed problem analysis, literature review, system architecture design, and module planning. The feasibility of integrating identification systems, IoT monitoring, and cloud storage into a single healthcare platform has been validated through research and system design.

In the next phase, the project will focus on hardware implementation, sensor integration, database connectivity, and prototype development. The LifeTag system has the potential to support faster medical decision-making, improve patient safety, and enhance emergency healthcare response, particularly in resource-limited environments.

Overall, this project highlights how interdisciplinary technologies can be combined to create practical healthcare solutions that address real-world problems.

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